

Question 1: Convolution and Correlation ($\Sigma = 12$)

- (a) How is the **convolution** $G = F * I$ defined, given an image I and a filter F ? Give the (___/2) formula.

- (b) State if the following properties of **correlation filtering** ($F \otimes H$) are always valid. (___/1)

i. $(a \cdot H + J) \otimes F = a \cdot H \otimes F + J \otimes F$

i. _____

ii. $F[i, j] \otimes H[i, j] = F[i, j] * H[-i, -j]$

ii. _____

- (c) What problems do occur at the border of the image when performing filtering? How (___/2) can these be avoided besides zero-padding? Describe two methods and name one advantage and one disadvantage for each of them.

- (d) Image filtering and the Fourier transform (___/2)

- i. The Fourier transform of a Gaussian is also a Gaussian.

i. _____

- ii. The Fourier transform of a box filter is also a box filter.

ii. _____

- iii. The Gaussian filter emphasizes the low frequencies.

iii. _____

- iv. Truncating the Fourier spectrum at a given frequency below the maximum frequency and only keeping the lower frequencies causes aliasing artifacts in the image.

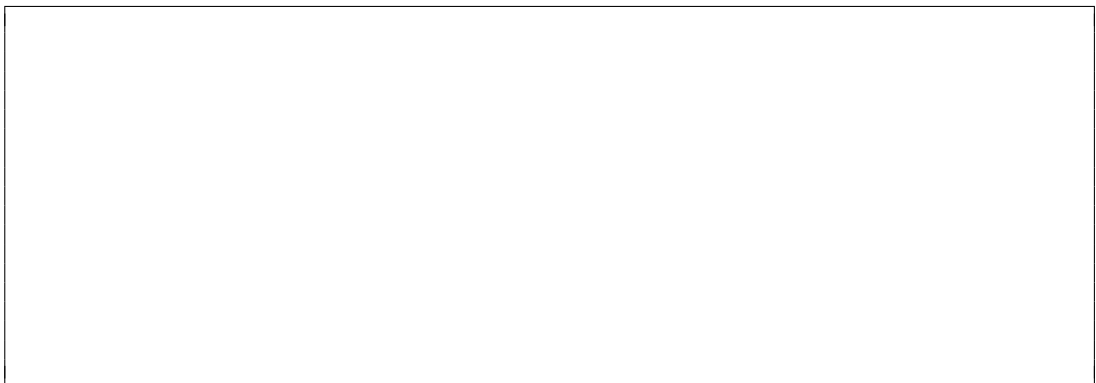
iv. _____

(e) What is a Gaussian Pyramid and how is it created?

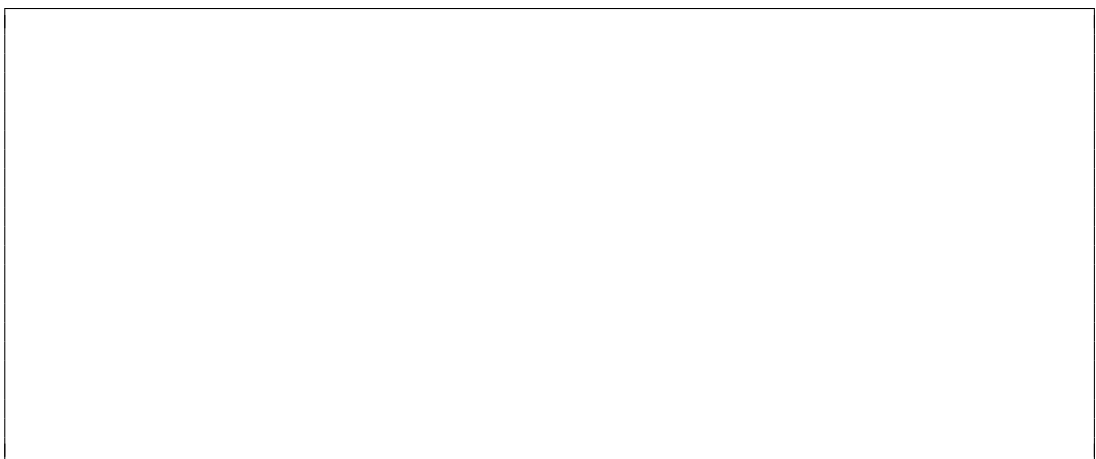
(___/2)



(f) Why is the Gaussian used for creating the pyramid? What would happen without this step? (___/1)



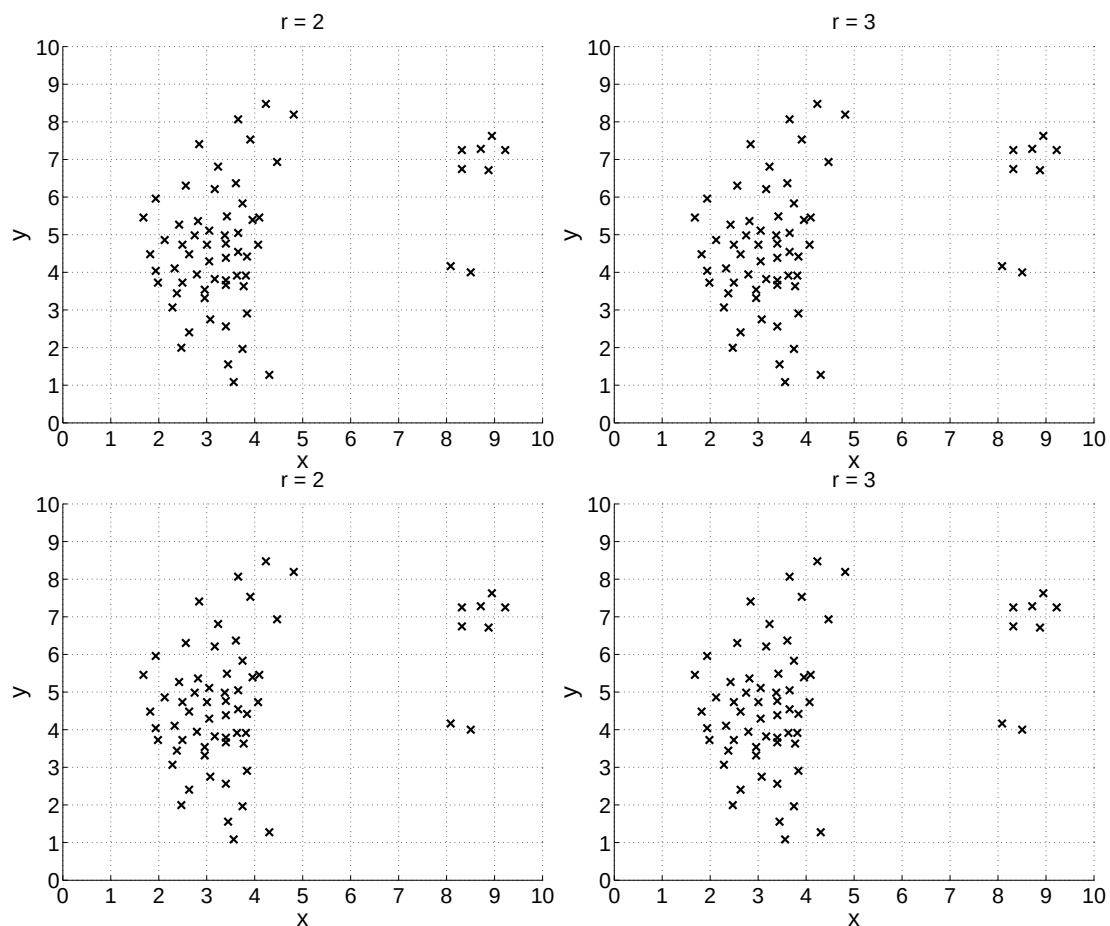
(g) What is the Laplacian Pyramid? And how is it connected to the Gaussian Pyramid? (___/2)



Question 2: Mean-Shift ($\Sigma = 11$)

(a) List the steps of the Mean-Shift algorithm.

(____/4)

(b) Sketch the (approximate) cluster boundaries Mean Shift would give for the following (____/2) dataset, for $r = 2$ (left) and $r = 3$ (right).

(c) Briefly describe a strategy to speed up Mean-Shift.

(___/2)

(d) List two advantages and two disadvantages of Mean-Shift.

(___/2)

(e) Briefly describe how Mean-Shift can be used for image segmentation.

(___/1)

Question 3: k-Means ($\Sigma = 16$)

- (a) List the steps of the k-Means algorithm.

(___/4)

- (b) Properties of k-Means

(___/3)

Will k-Means always converge?

☐ Yes ☐ No

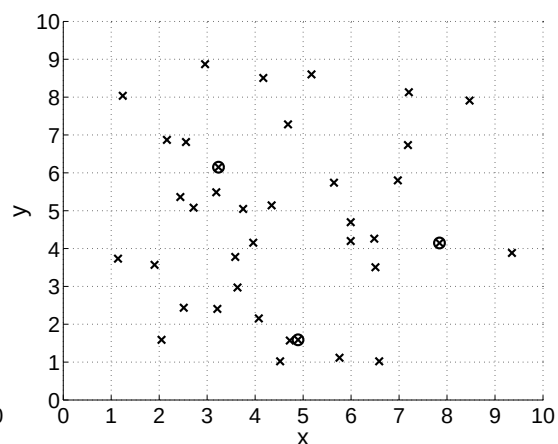
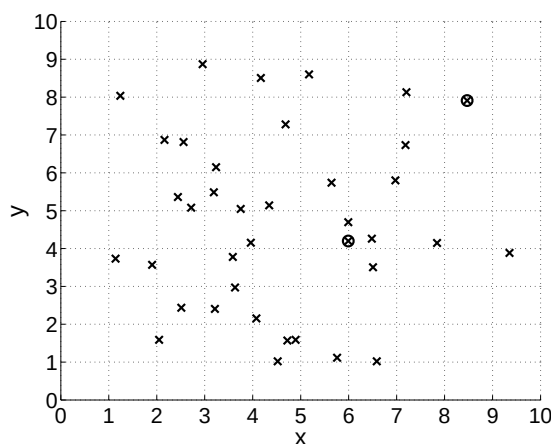
Does k-Means always find the best solution with respect to its objective function?

☐ Yes ☐ No

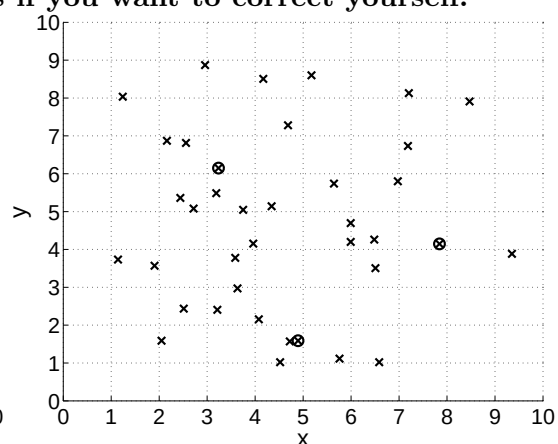
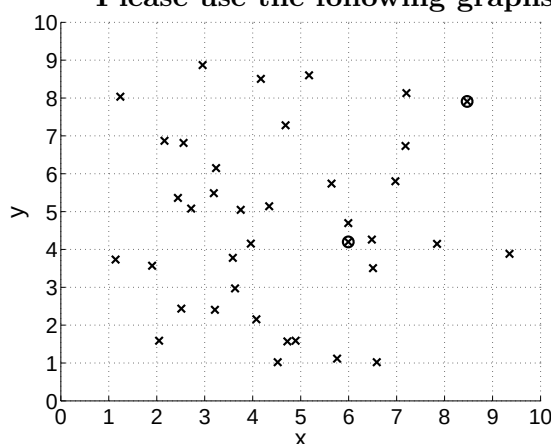
Is the problem of finding the optimal solution NP-complete?

☐ Yes ☐ No

- (c) Sketch the (approximate) cluster boundaries and their means k-Means would give for the following dataset for
- $k = 2$
- (left) and
- $k = 3$
- (right). The circled points are the initial means. (___/4)



Please use the following graphs if you want to correct yourself.



(d) List one advantage and two disadvantages of k-Means.

(___/3)

(e) Describe in detail how k-Means can be used for image segmentation.

(___/2)

Question 4: Local Features ($\Sigma = 12$)

- (a) Please fill in the following Matlab code fragment to complete the Hessian detector. (___/4)
(Pseudo-code is sufficient, as long it is unambiguously clear what is meant.)

```

1 function [px, py] = computeHessian(filename, sigma, thresh)
2   % -----
3   % Preprocessing
4   % -----
5   I          = loadImage(filename);
6   Ig         = gaussianfilter(I, sigma); % Gaussian filter
7   [Ix, Iy]    = gaussderiv(I, sigma);    % first derivatives
8   [Ixx, Ixy, Iyy] = gaussderiv2(I, sigma); % second derivatives
9
10  % -----
11  % Compute Hessian score for each pixel
12  % -----
13  [height, width] = size(I);
14  score = zeros(height, width);
15  for y = 1:height
16    for x = 1:width
17      % Compute Hessian score for pixel I(y, x) and store it in score(y,
        x)

```

```

18  end
19  end
20
21  % -----
22  % Postprocessing
23  % -----
24  % Extract the interest points from the computed score map.
25  [py, px] = find(score > thresh)
26 end

```

- (b) The above code is still not fully correct. There are 2 steps missing. Please point them (___/2)
out (A verbal explanation is sufficient).

(c) Details to the Hessian detector.

- i. What are the differences between the Hessian and the Harris detector? Please (___/2) briefly describe.

- ii. Properties of Hessian keypoints (___/2)

Is the Hessian detector scale-invariant?

☐ Yes ☐ No

Is the Hessian detector translation-invariant?

☐ Yes ☐ No

- iii. How can Harris be extended to detect key points with arbitrary scale automatically. (___/2)

Question 5: Viola-Jones ($\Sigma = 15$)(a) The Adaboost *training* algorithm.

i. What is the input and what is the output of this algorithm?

(____/2)

ii. Briefly explain the steps of the Adaboost *training* algorithm (no formulas required). (____/3)

(b) Which property has to be fulfilled by the weak classifiers?

(____/1)

(c) How is a test point classified? Give the equation.

(____/2)

- (d) What are the weak classifiers that are used for Viola-Jones face detection? (You may (___/2) sketch to support your answer.)

- (e) Integral Images

- i. What is an integral image?

(___/1)

- ii. Why and how are integral images used for Viola-Jones face detection? (You may (___/2) sketch to support your answer.)

- (f) Briefly explain how cascading classifiers for detection works.

(___/2)

Question 6: Fundamental Matrix ($\Sigma = 15$)

(a) Briefly explain the following concepts

i. Essential Matrix

(____/1)

ii. Fundamental Matrix

(____/1)

iii. Epipolar line

(____/1)

iv. Epipolar plane

(____/1)

(b) Eight-point algorithm

(____/4)

- i. Fill in the first row of the following matrix in order to complete the Eight-point algorithm. Assume that the point correspondence is called $(\mathbf{x}, \mathbf{y})$ where $\mathbf{x} = (x_1, x_2, 1)$ is located in the left image and $\mathbf{y} = (y_1, y_2, 1)$ in the right image. (**Hint:** Try to write out the epipolar constraint).

$$\begin{bmatrix} \square & \square & \square & \square & \square & \square & \square & \square & \square \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \cdot \begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{bmatrix} = 0$$

ii. How do we solve this equation? What exactly is the solution? (___/1)

- iii. Solving this equation directly usually leads to very inaccurate results in presence (___/1) of noise (even without outlier). Please explain why.

- iv. What can we do to about this issue in order to get more accurate results, and (___/1) how?

(c) Rank constraints of the Fundamental Matrix.

- i. What is the rank of the Fundamental matrix? Why?

(___/2)

- ii. What would happen if F had full rank?

(___/1)

- iii. How do we enforce the rank constraint when estimating F ?

(___/1)